
Data Analytics Internship

Name: Akshata.T.Rathod

USN: 4JN22CS011

College: Jawaharlal Nehru New College of Engineering

Branch: Computer Science and Engineering

Internship Company: Rooman Technologies

Domain: Data Analytics

Project: Business Insights Dashboard using Superstore Dataset

Mail: akshata.akshu.2004@gmail.com

Date: 08-05-2026

PHASE 3 - PRESENT

Project Title

Business Insights Dashboard using Superstore Dataset

Project Type

Individual Project

1. Introduction

This project focuses on analyzing retail sales data to extract meaningful insights and support better decision-making. Using the Superstore dataset, the project applies data analysis and visualization techniques to understand business performance across different dimensions such as category, region, and time.

The project was carried out in three phases: planning (Blueprint), implementation (Build), and presentation (Final Phase).

2. Problem Addressed

Businesses generate large volumes of data, but it is often difficult to interpret and extract useful insights from it. Traditional reports fail to clearly highlight trends, performance gaps, and loss areas.

This project addresses this challenge by providing clear visual insights, identifying key performance drivers, and highlighting factors that impact profitability, such as discount strategies and regional performance.

3. Approach Followed

The project followed a structured approach:

-
- Data cleaning and preprocessing using Python
 - Exploratory Data Analysis (EDA) to understand patterns
 - Creation of visualizations to identify trends and relationships
 - Development of dashboards using Power BI and React
 - Interpretation of insights and validation using correlation analysis

4. Key Results

The analysis produced several important findings:

- Technology category is the highest revenue-generating category
- West region shows the best performance in terms of sales and profit
- Higher discounts negatively impact profitability and may lead to losses
- A small number of products contribute significantly to total revenue
- Sales vary over time, indicating trends and possible seasonality

These results help in understanding business performance and identifying areas for improvement.

5. Solution and Contribution

The project delivers a complete data analytics solution through:

- A Jupyter Notebook for data analysis and visualization
- A Power BI dashboard for interactive business insights
- A React-based dashboard for advanced and flexible analysis

These tools make it easier to explore data, understand trends, and support decision-making.

6. Impact of the Project

This project demonstrates how data analytics can be applied in real-world business scenarios:

- Helps organizations identify profitable and non-profitable areas
- Supports better strategic and operational decision-making
- Improves understanding of customer and product performance
- Reduces business risk by identifying loss factors such as high discounts

7. Learning Outcomes

Through this project, the following skills were developed:

- Data cleaning and preprocessing techniques
- Exploratory Data Analysis (EDA)
- Data visualization and interpretation
- Dashboard development using Power BI
- Frontend dashboard development using React
- Understanding of business analytics concepts

8. Challenges Faced

- Handling and preparing raw data for analysis
- Selecting appropriate visualizations for clear insights
- Interpreting relationships between variables
- Designing dashboards for better usability

These challenges helped improve problem-solving and analytical thinking skills.

9. Future Scope

- Integration of real-time data for continuous analysis
- Application of predictive analytics for forecasting
- Enhancement of dashboard features and interactivity
- Use of machine learning models for deeper insights

10. Conclusion

This project successfully transforms raw data into meaningful business insights through analysis and visualization. It demonstrates how data analytics can be used to identify trends, improve performance, and support data-driven decision-making.

Overall, the project highlights the importance of analytics in solving real-world business problems and improving organizational outcomes.

11. Project Links

- **GitHub Repository (Notebook + Power BI):**
<https://github.com/akshata130504/Business-Insights-Dashboard-using-Superstore-Dataset>
- **React Dashboard Repository:**
<https://github.com/akshata130504/Business-Insights-Dashboard-using-React->
- **React Dashboard Live Link:**
<https://business-insights-dashboa-git-43abc8-akshata-t-rathods-projects.vercel.app>
- **Presentation (PPT):**
<https://canva.link/kbxo3aujq7pk9pm>